

TAYLOR PENTZ

Senior Software Engineer

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Professional Summary

Innovative and impact-driven Senior Software Engineer with 9+ years of full-stack experience in startups and high-growth environments. Proven expertise in designing and delivering scalable, real-time platforms across finance and SaaS. Skilled in Python, TypeScript, and React, with deep knowledge of cloud-native infrastructure (AWS, GCP), DevOps, and CI/CD. Passionate about building products from zero-to-one, architecting resilient systems, and working on teams that value fast iteration and technical ownership.

Technical Skills

- ❖ **Languages & Frameworks:** Python, TypeScript, JavaScript, React, Node.js, Next.js, FastAPI, Express, C#
- ❖ **Cloud & DevOps:** AWS (Lambda, S3, RDS, ECS), GCP (Firestore, Cloud Functions), Docker, GitHub Actions, Terraform
- ❖ **Databases:** PostgreSQL, MySQL, MongoDB, Redis, DynamoDB, Firestore
- ❖ **Architecture:** Microservices, Event-Driven Systems, Serverless, CQRS, REST & GraphQL APIs
- ❖ **Tools:** Kafka, RabbitMQ, WebSockets, Git, ELK Stack, Prometheus, Cypress, Jenkins
- ❖ **Soft Skills:** Technical Leadership, Mentoring, Agile/Scrum, Cross-Functional Collaboration

Professional Experience

SumatoSoft

Senior Software Engineer

Boston, MA (Remote) | Apr 2023 – Mar 2025

- Spearheaded the architecture and development of real-time microservices using Python FastAPI and PostgreSQL, improving financial data ingestion speed by 300% while maintaining accuracy at scale.
- Designed and implemented investment workflow orchestration logic integrated with third-party APIs for automated reconciliation, audit trails, and ledger consistency in high-throughput environments.
- Deployed scalable frontend solutions using React and Next.js, enabling millisecond refresh rates for live portfolio dashboards with WebSocket-based push updates.
- Led cross-functional sprints, mentoring 4+ junior engineers and coordinating product releases from ideation to post-release retrospectives across distributed teams.
- Introduced test-driven development culture and CI/CD pipelines with GitHub Actions and Docker, reducing bug reports in production by 65% over two quarters.
- Integrated cloud-native solutions with AWS services including Lambda, S3, RDS, and API Gateway to build highly available RESTful and event-driven systems.
- Contributed to internal SDKs and developer tooling libraries, reducing API integration time for new investment products by 45%.
- Actively collaborated with product stakeholders to define roadmap features, aligning engineering priorities with investor and compliance requirements.

Hexaview Technologies Inc.

Software Engineer

New York, NY | Jul 2018 – Mar 2023

- Delivered end-to-end wealth management dashboards in a startup fintech setting, integrating Plaid and Yodlee APIs to power seamless account aggregation.
- Built complex investment simulation engines using Node.js and TypeScript, allowing advisors to generate scenario-based performance reports for clients.
- Led migration of legacy PHP backend to Python microservices, reducing deployment times and maintenance overhead by 70%.
- Introduced Apache Kafka for asynchronous trade processing, resulting in 80% throughput improvement in reconciliation tasks.
- Managed AWS ECS container orchestration and Terraform-based IaC to support scalable deployments across dev/staging/prod environments.
- Built automated trade compliance checks for regulatory enforcement using a rule-based validation engine with PostgreSQL triggers and stored procedures.
- Collaborated with design and product teams to develop reusable React component libraries aligned with WCAG accessibility standards.
- Pioneered feature flag-based release processes to enable safe rollout of beta features and A/B testing strategies in production.

SimbirSoft Company

Software Developer

Boston, MA | Mar 2015 – Jun 2018

- Developed high-throughput B2B SaaS tools for managing financial operations, supporting over 10,000 concurrent users at peak traffic.
- Led the creation of an internal permissions management microservice using JWT and OAuth2 standards, improving system-wide security compliance.
- Built internal dashboards with D3.js and React for financial analysts to monitor KPIs and detect anomalies in transactional data.
- Enhanced logging and observability using ELK stack and custom middleware, reducing incident detection time from hours to minutes.
- Maintained robust REST and GraphQL APIs using Express.js and TypeScript, achieving 99.98% uptime across 12-month rolling periods.
- Collaborated with QA engineers to implement BDD frameworks using Cucumber.js and Cypress for critical user flows.
- Refactored legacy reporting system to a distributed architecture, enabling dynamic report generation and export at scale.
- Participated in sprint planning, grooming, and team retrospectives as part of an Agile Scrum team with continuous delivery targets.

Microsoft

Software Developer Intern

Cambridge, MA | Aug 2014 – Feb 2015

- Supported early-stage development of cloud-native developer tools on Azure platform, contributing to backend API enhancements in C#.
- Built internal diagnostics dashboard for engineering leadership to monitor service health and feature adoption metrics.
- Automated testing pipelines using Jenkins and NUnit, reducing regression-related downtime by 30%.
- Conducted code reviews and implemented performance profiling tools to optimize latency in developer workflows.
- Integrated Git for version control and trained junior interns on Git branching strategies for large-team collaboration.
- Attended weekly architecture meetings and contributed to RFCs for new service components.
- Participated in hackathons and innovation days; co-developed a plugin system that became part of internal toolchain.
- Documented internal tooling and authored onboarding guides adopted across engineering teams.

Projects

- ✓ **Real-Time Investment Platform** (*SumatoSoft, 2024*)
Built a real-time investment platform for retail users, handling live asset pricing, order execution, and portfolio tracking with millisecond-level accuracy. Enabled seamless onboarding, trade execution, and automated compliance checks.
- ✓ **Event-Driven Wealth Management System** (*Hexaview, 2021*)
Led backend design of a Kafka-based portfolio alerting system that processed 5M+ events daily. Implemented rules engine for proactive client engagement and risk management.

Certifications

- AWS Certified Solutions Architect – Associate
- Google Associate Cloud Engineer
- MongoDB for Developers (MongoDB University)
- Python for Data Structures and Algorithms (Coursera)

Education

University of North Carolina at Chapel Hill

Bachelor's Degree in Computer Science (2010 – 2014)